Paper 26 — General normalization law for the graph Laplacian

$W_{ij} = \exp[-d_g(x_i, x_j)^2/(4\varepsilon)]$
local Gaussian graph

→

$L_{N,\varepsilon} = D - W$
unnormalized Laplacian

→

$c_{N,\varepsilon} L_{N,\varepsilon} \to -\Delta_g$
continuum limit

$$c_{N,\varepsilon} = \frac{1}{\rho(4\pi)^{D/2}\varepsilon^{D/2+1}}, \quad \rho = \frac{N}{\mathrm{Vol}(M)}$$

$S^1: \ c = \frac{\sqrt{\pi}}{N\varepsilon^{3/2}}$

$T^2: \ c = \frac{\pi}{N\varepsilon^2}$

$S^2: \ c = \frac{1}{N\varepsilon^2}$